

Name: Aashu Tiwari

PRN: 125M1H051

Assignment No: 8

Task 1: Assignments based on Javascript Basics - Part 1

1) Write a JavaScript program to:

- Accept marks of 5 students using input fields.
- Store them in an array.
- Display:
 - Total marks
 - Average marks
 - Highest marks

```
<!DOCTYPE html>
<html>
<head><title>Student Marks</title></head>
<body>
<h2>Student Marks Calculator</h2>
<input type="number" id="m1" placeholder="Student 1 Marks"><br><br>
<input type="number" id="m2" placeholder="Student 2 Marks"><br><br>
<input type="number" id="m3" placeholder="Student 3 Marks"><br><br>
<input type="number" id="m4" placeholder="Student 4 Marks"><br><br>
<input type="number" id="m5" placeholder="Student 5 Marks"><br><br>
<button onclick="calculate()">Calculate</button>
<h3 id="result"></h3>
<script>
function calculate() {
    let marks = [
        Number(document.getElementById("m1").value),
        Number(document.getElementById("m2").value),
        Number(document.getElementById("m3").value),
        Number(document.getElementById("m4").value),
        Number(document.getElementById("m5").value)
    ];
    let total = marks.reduce((a,b)=>a+b,0);
    let average = total / marks.length;
    let highest = Math.max(...marks);
```

```
document.getElementById("result").innerHTML =  
    "Total Marks: " + total + "<br>" +  
    "Average Marks: " + average + "<br>" +  
    "Highest Marks: " + highest;  
}  
</script>  
</body>  
</html>
```

OUTPUT:

Student Marks

127.0.0.1:5500/Task1/student_marks.html

Student Marks Calculator

Student 1 Marks

Student 2 Marks

Student 3 Marks

Student 4 Marks

Student 5 Marks

Student Marks

127.0.0.1:5500/Task1/student_marks.html

Student Marks Calculator

25

45

78

23

54

Total Marks: 225
Average Marks: 45
Highest Marks: 78

2) Create a program that:

- Accepts 10 numbers from the user.
- Stores them in an array.
- Separates and displays even and odd numbers in two different lists.

```
• <!DOCTYPE html>
• <html>
• <head><title>Even Odd Separator</title></head>
• <body>
• <h2>Even and Odd Number Separator</h2>
• <textarea id="numbers" placeholder="Enter numbers separated by
  comma"></textarea><br><br>
• <button onclick="separate()">Separate</button>
• <h3 id="output"></h3>
•
• <script>
• function separate() {
•   let arr =
document.getElementById("numbers").value.split(",").map(Number);
•   let even = [];
•   let odd = [];
•
•   for(let num of arr) {
•     if(num % 2 === 0) even.push(num);
•     else odd.push(num);
•   }
•
•   document.getElementById("output").innerHTML =
•     "Even Numbers: " + even.join(", ") + "<br>" +
•     "Odd Numbers: " + odd.join(", ");
• }
• </script>
• </body>
• </html>
```

OUTPUT:

Even Odd Separator

Ask Gemini

127.0.0.1:5500/task1/even_odd_separator.html

School

Even and Odd Number Separator

Enter numbers separated by comma

Separate

Even Odd Separator

Ask Gemini

127.0.0.1:5500/task1/even_odd_separator.html

School

Even and Odd Number Separator

5, 7, 2, 3, 8, 0, 1, 56, 8, 2, 45

Separate

Even Numbers: 2, 8, 0, 56, 8, 2

Odd Numbers: 5, 7, 3, 1, 45

3) Develop a program to:

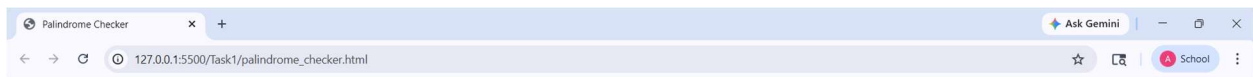
- Accept a string from the user.
- Check whether it is a palindrome.
- Display appropriate message.

```
<!DOCTYPE html>
<html>
<head><title>Palindrome Checker</title></head>
<body>
<h2>Palindrome Checker</h2>
<input type="text" id="text" placeholder="Enter text">
<button onclick="checkPalindrome()">Check</button>
<h3 id="result"></h3>

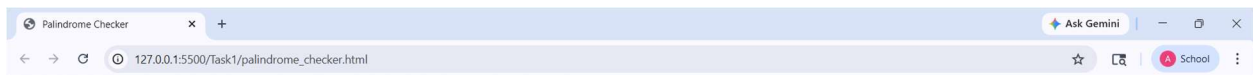
<script>
function checkPalindrome() {
    let str = document.getElementById("text").value.toLowerCase();
    let reversed = str.split("").reverse().join("");

    if(str === reversed)
        document.getElementById("result").innerHTML = "It is a Palindrome";
    else
        document.getElementById("result").innerHTML = "It is not a Palindrome";
}
</script>
</body>
</html>
```

OUTPUT:



Palindrome Checker



Palindrome Checker

It is a Palindrome

4) Write a JavaScript program that:

- Takes a sentence as input.
- Counts number of words, characters (excluding spaces), and vowels.

```
• <!DOCTYPE html>  
• <html>  
• <head><title>Sentence Analyzer</title></head>
```

```

• <body>
• <h2>Sentence Analyzer</h2>
• <textarea id="sentence"></textarea><br><br>
• <button onclick="analyze()">Analyze</button>
• <h3 id="result"></h3>
•
• <script>
• function analyze() {
•     let sentence = document.getElementById("sentence").value;
•     let words = sentence.trim().split(/\s+/).length;
•     let characters = sentence.replace(/\s/g, "").length;
•     let vowels = sentence.match(/[aeiouAEIOU]/g);
•     let vowelCount = vowels ? vowels.length : 0;
•
•     document.getElementById("result").innerHTML =
•         "Words: " + words + "<br>" +
•         "Characters: " + characters + "<br>" +
•         "Vowels: " + vowelCount;
• }
• </script>
• </body>
• </html>

```

OUTPUT:





5) Create a program to:

- Accept user's birthdate.
- Calculate and display the exact age in years, months, and days using JavaScript Date object.

```
• <!DOCTYPE html>
• <html>
• <head><title>Age Calculator</title></head>
• <body>
• <h2>Age Calculator</h2>
• <input type="date" id="birthdate">
• <button onclick="calculateAge()">Calculate</button>
• <h3 id="result"></h3>
•
• <script>
• function calculateAge() {
•     let birthDate = new Date(document.getElementById("birthdate").value);
•     let today = new Date();
•
•     let years = today.getFullYear() - birthDate.getFullYear();
•     let months = today.getMonth() - birthDate.getMonth();
•     let days = today.getDate() - birthDate.getDate();
•
•     if(days < 0) {
•         months--;
```



```
•     days += 30;
•   }
•
•   if(months < 0) {
•     years--;
•     months += 12;
•   }
•
•   document.getElementById("result").innerHTML =
•     years + " Years " + months + " Months " + days + " Days";
• }
• </script>
• </body>
• </html>
```

OUTPUT:



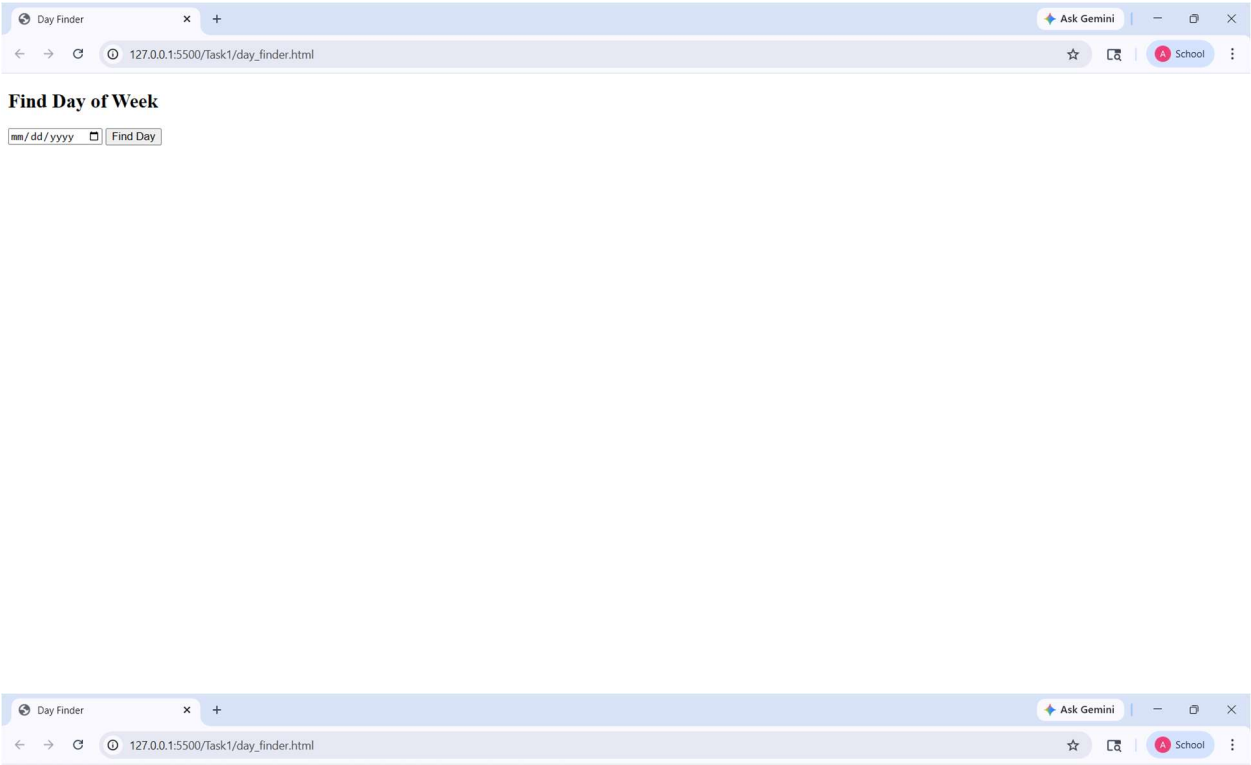


6) Write a program that:

- Accepts a date input.
- Displays the day of the week (e.g., Monday, Tuesday).

```
• <!DOCTYPE html>
• <html>
• <head><title>Day Finder</title></head>
• <body>
• <h2>Find Day of Week</h2>
• <input type="date" id="date">
• <button onclick="findDay()">Find Day</button>
• <h3 id="result"></h3>
•
• <script>
• function findDay() {
•     let inputDate = new Date(document.getElementById("date").value);
•     let days =
• ["Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"];
•     document.getElementById("result").innerHTML =
•         "Day: " + days[inputDate.getDay()];
• }
• </script>
• </body>
• </html>
```

OUTPUT:



Find Day of Week

12/05/2026 Find Day

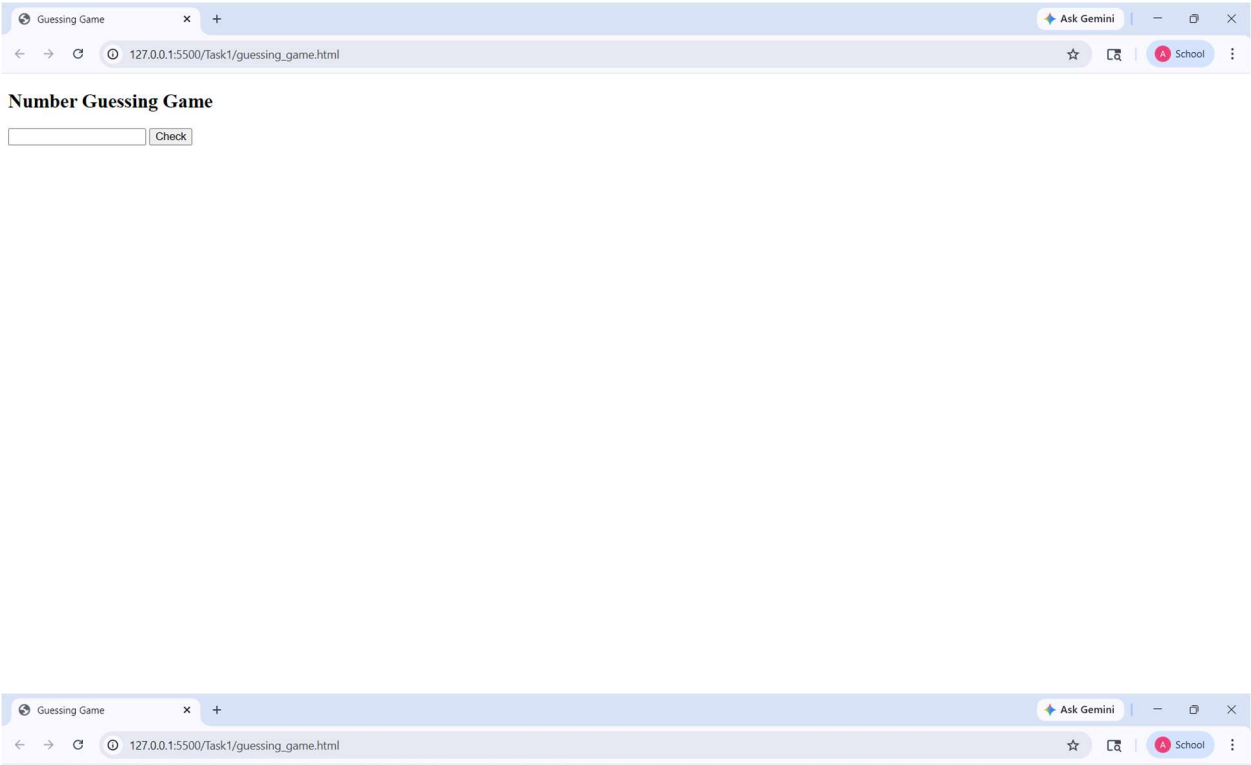
Day: Saturday

7) Develop a number guessing game:

- Generate a random number between 1–100 using a Math object.
- Allow users to guess the number.
- Display whether the guess is too high, too low, or correct.

```
• <!DOCTYPE html>
• <html>
• <head><title>Guessing Game</title></head>
• <body>
• <h2>Number Guessing Game</h2>
• <input type="number" id="guess">
• <button onclick="checkGuess()">Check</button>
• <h3 id="result"></h3>
•
• <script>
• let randomNumber = Math.floor(Math.random() * 100) + 1;
•
• function checkGuess() {
•     let guess = Number(document.getElementById("guess").value);
•
•     if(guess > randomNumber)
•         document.getElementById("result").innerHTML = "Too High!";
•     else if(guess < randomNumber)
•         document.getElementById("result").innerHTML = "Too Low!";
•     else
•         document.getElementById("result").innerHTML = "Correct Guess!";
• }
• </script>
• </body>
• </html>
```

OUTPUT:



Number Guessing Game

10

Too Low!

Number Guessing Game

50

Check

Too High!

Number Guessing Game

35

Check

Correct Guess!

8) Create a calculator that:

- Accepts two numbers and an operator (+, -, *, /).
- Performs calculation using Math functions.
- Displays the result.

```
• <!DOCTYPE html>
• <html>
• <head><title>Calculator</title></head>
• <body>
• <h2>Simple Calculator</h2>
• <input type="number" id="num1"><br><br>
• <input type="number" id="num2"><br><br>
•
• <select id="operator">
• <option value="+">+</option>
• <option value="-">-</option>
• <option value="*">*</option>
• <option value="/">/</option>
• </select>
•
• <button onclick="calculate()">Calculate</button>
• <h3 id="result"></h3>
•
• <script>
• function calculate() {
•     let num1 = Number(document.getElementById("num1").value);
•     let num2 = Number(document.getElementById("num2").value);
•     let operator = document.getElementById("operator").value;
•     let result;
•
•     switch(operator) {
•         case "+": result = num1 + num2; break;
•         case "-": result = num1 - num2; break;
•         case "*": result = num1 * num2; break;
•         case "/": result = num1 / num2; break;
•     }
•
•     document.getElementById("result").innerHTML = "Result: " + result;
• }
• </script>
• </body>
• </html>
```

OUTPUT:

Calculator

Ask Gemini

127.0.0.1:5500/Task1/calculator.html

School

Simple Calculator

5

0

*

Calculate

Result: 0

Calculator

Ask Gemini

127.0.0.1:5500/Task1/calculator.html

School

Simple Calculator

4567

5677

-

Calculate

Result: -1110

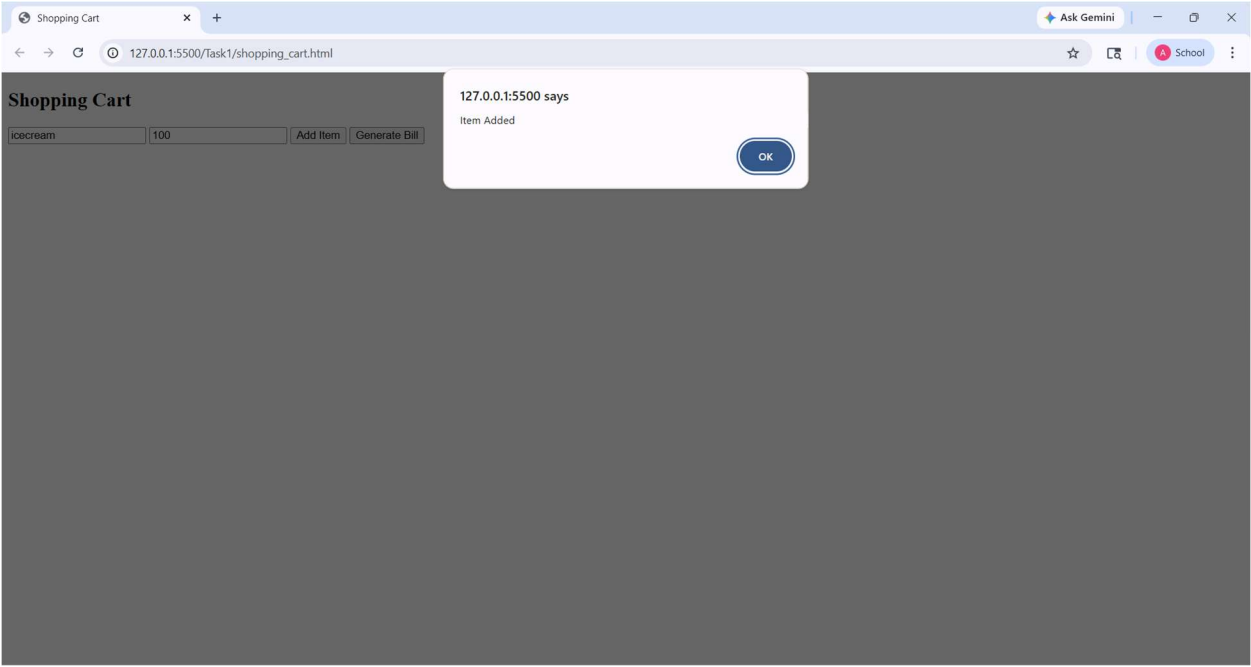
9) Build a simple shopping cart:

- Accept item names and prices.
- Store them in arrays.
- Display:
 - List of items
 - Total bill
 - Apply 10% discount if total > ₹1000

```
• <!DOCTYPE html>
• <html>
• <head><title>Shopping Cart</title></head>
• <body>
• <h2>Shopping Cart</h2>
•
• <input type="text" id="item" placeholder="Item Name">
• <input type="number" id="price" placeholder="Price">
•
• <button onclick="addItem()">Add Item</button>
• <button onclick="showBill()">Generate Bill</button>
•
• <h3 id="result"></h3>
•
• <script>
• let items = [];
• let prices = [];
•
• function addItem() {
•   items.push(document.getElementById("item").value);
•   prices.push(Number(document.getElementById("price").value));
•   alert("Item Added");
• }
•
• function showBill() {
•   let total = prices.reduce((a,b)=>a+b,0);
•   let discount = total > 1000 ? total * 0.10 : 0;
•   let finalAmount = total - discount;
•
•   document.getElementById("result").innerHTML =
•     "Items: " + items.join(", ") + "<br>" +
•     "Total: ₹" + total + "<br>" +
•     "Discount: ₹" + discount + "<br>" +
•     "Final Amount: ₹" + finalAmount;
• }
• </script>
```

- `</body>`
- `</html>`

OUTPUT:



Task 2: Assignments based on Javascript Basics - Part 2

1) Design a program that:

- Accepts a list of log messages (array of strings).
- Each log contains: "ERROR: file not found at 10:30AM"
- Tasks:
 - Count occurrences of each log type (ERROR, WARNING, INFO)
 - Extract time and sort logs chronologically
 - Display summary report

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-
scale=1.0">
•   <title>Log Message Analyzer</title>
•   <style>
•     body {
•       font-family: Arial, sans-serif;
•       background: #f4f4f4;
•       padding: 20px;
•     }
•
•     .container {
•       max-width: 800px;
•       margin: auto;
•       background: white;
•       padding: 20px;
•       border-radius: 10px;
•       box-shadow: 0 0 10px rgba(0,0,0,0.1);
•     }
•
•     textarea {
•       width: 100%;
•       height: 180px;
•       padding: 10px;
•       font-size: 16px;
•     }
•   </style>
```

```

    button {
        margin-top: 10px;
        padding: 10px 20px;
        background: #007bff;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
    }

    button:hover {
        background: #0056b3;
    }

    #output {
        margin-top: 20px;
        background: #f9f9f9;
        padding: 15px;
        border-radius: 5px;
    }
</style>
</head>
<body>
    <div class="container">
        <h2>Log Message Analyzer</h2>

        <p>Enter one log per line:</p>

        <textarea id="logs">
ERROR: file not found at 10:30AM
INFO: server started at 09:15AM
WARNING: memory usage high at 11:00AM
ERROR: database failed at 08:45AM
INFO: backup completed at 01:20PM
        </textarea>

        <button onclick="analyzeLogs()">Analyze Logs</button>

        <div id="output"></div>
    </div>

    <script>
        function convertToMinutes(timeStr) {
            let match = timeStr.match(/(\d+):(\d+)(AM|PM)/);

```

```

    let hours = parseInt(match[1]);
    let minutes = parseInt(match[2]);
    let period = match[3];

    if (period === 'PM' && hours !== 12) {
        hours += 12;
    }

    if (period === 'AM' && hours === 12) {
        hours = 0;
    }

    return hours * 60 + minutes;
}

function analyzeLogs() {
    let logText = document.getElementById('logs').value.trim();
    let logs = logText.split('\n');

    let counts = {
        ERROR: 0,
        WARNING: 0,
        INFO: 0
    };

    let logObjects = [];

    logs.forEach(log => {
        let type = log.split(':')[0];
        counts[type]++;

        let timeMatch = log.match(/\d{1,2}:\d{2}(AM|PM)/);
        let time = timeMatch ? timeMatch[0] : '00:00AM';

        logObjects.push({
            text: log,
            time: time,
            minutes: convertToMinutes(time)
        });
    });

    logObjects.sort((a, b) => a.minutes - b.minutes);

    let output = `

```

```

    <h3>Summary Report</h3>
    <p><strong>ERROR:</strong> ${counts.ERROR}</p>
    <p><strong>WARNING:</strong> ${counts.WARNING}</p>
    <p><strong>INFO:</strong> ${counts.INFO}</p>

    <h3>Chronological Logs</h3>
    <ul>
    `;

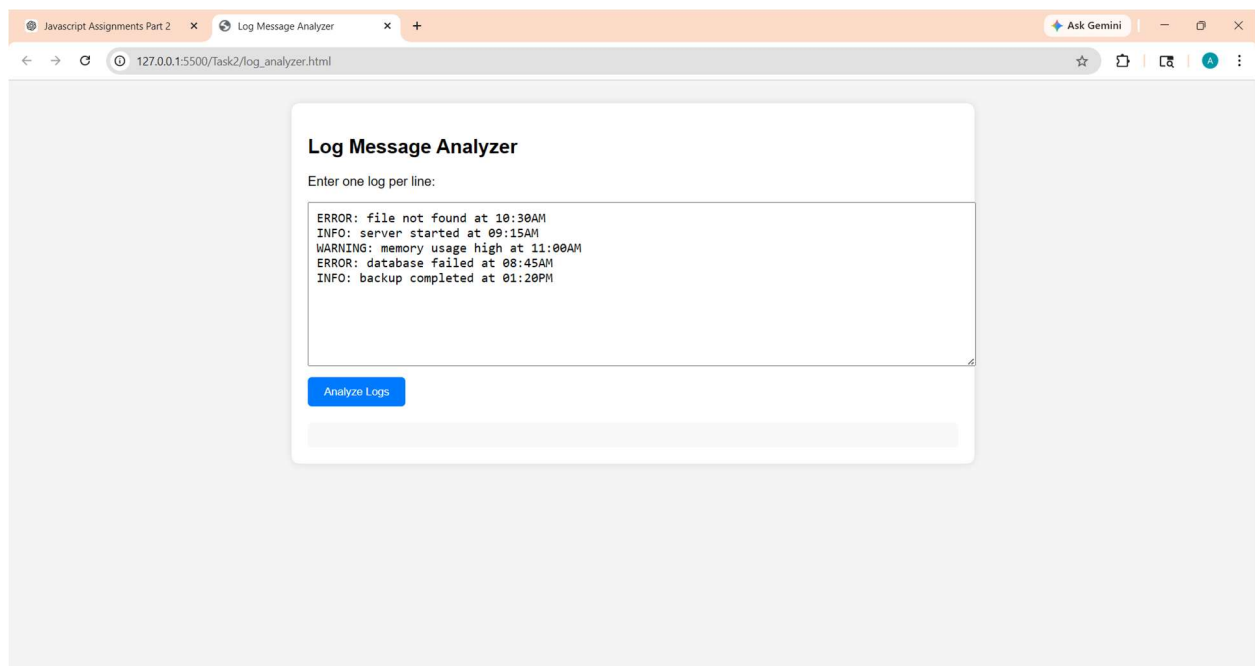
    logObjects.forEach(log => {
        output += `<li>${log.text}</li>`;
    });

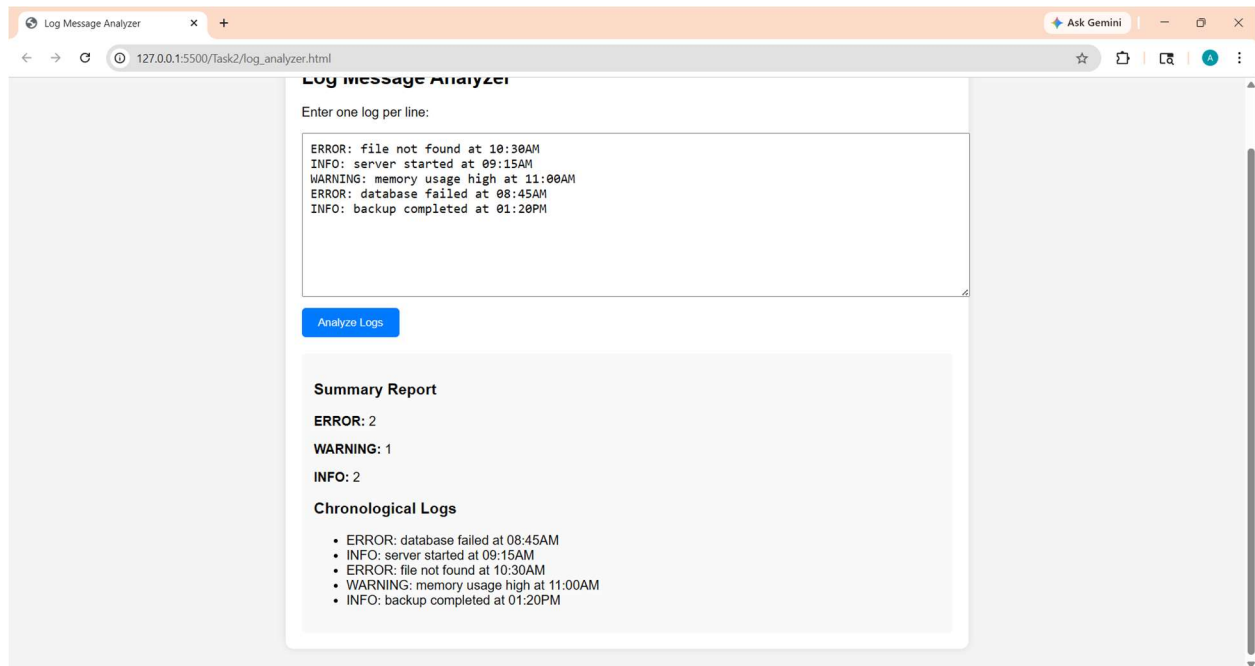
    output += '</ul>';

    document.getElementById('output').innerHTML = output;
    }
</script>
</body>
</html>

```

OUTPUT:





2) Create a system that:

- Accepts multiple future event dates.
- For each event:
 - Display countdown (days, hours, minutes)
 - Highlight overdue events
- Auto-update countdown every second

```

• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>Event Countdown</title>
•
•   <style>
•     body {
•       font-family: Arial, sans-serif;
•       background: #f0f0f0;
•       padding: 20px;
•     }
•
•     .container {
•       max-width: 700px;

```

```

    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 10px;
}

input {
    padding: 10px;
    margin: 5px;
}

button {
    padding: 10px 20px;
    background: green;
    color: white;
    border: none;
    cursor: pointer;
}

.event {
    margin-top: 15px;
    padding: 10px;
    border-radius: 5px;
    background: #f9f9f9;
}

.overdue {
    background: #ffcccc;
    color: red;
}
</style>
</head>
<body>

<div class="container">
    <h2>Future Event Countdown System</h2>

    <input type="text" id="eventName" placeholder="Event Name">
    <input type="datetime-local" id="eventDate">

    <button onclick="addEvent()">Add Event</button>

    <div id="eventList"></div>
</div>

```



```

<script>
  let events = [];

  function addEvent() {
    let name = document.getElementById('eventName').value;
    let date = document.getElementById('eventDate').value;

    if(name && date) {
      events.push({
        name: name,
        date: new Date(date)
      });

      document.getElementById('eventName').value = '';
      document.getElementById('eventDate').value = '';

      displayEvents();
    }
  }

  function displayEvents() {
    let container = document.getElementById('eventList');
    container.innerHTML = '';

    events.forEach(event => {
      let now = new Date();
      let diff = event.date - now;

      let div = document.createElement('div');
      div.classList.add('event');

      if(diff < 0) {
        div.classList.add('overdue');
        div.innerHTML = `
          <h3>${event.name}</h3>
          <p>Event Overdue</p>
        `;
      } else {
        let days = Math.floor(diff / (1000 * 60 * 60 * 24));
        let hours = Math.floor((diff / (1000 * 60 * 60)) % 24);
        let minutes = Math.floor((diff / (1000 * 60)) % 60);
        let seconds = Math.floor((diff / 1000) % 60);

        div.innerHTML = `

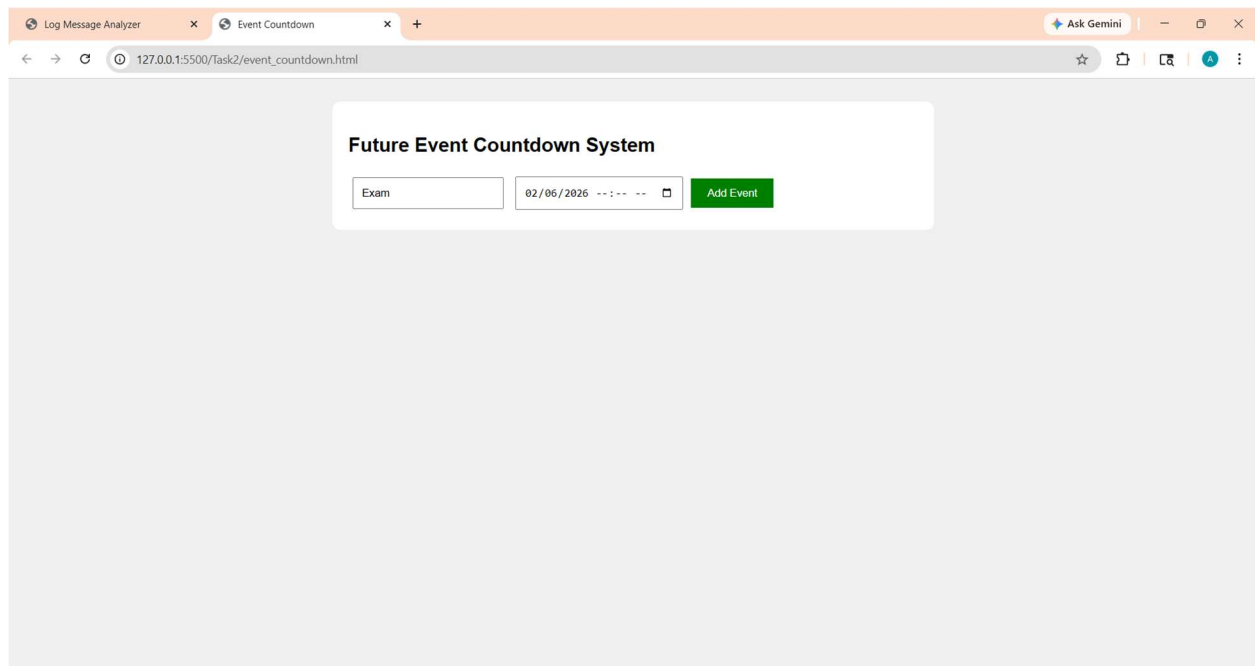
```

```

•         <h3>${event.name}</h3>
•         <p>${days} Days ${hours} Hours ${minutes} Minutes
    ${seconds} Seconds</p>
•         `;
•     }
•
•     container.appendChild(div);
•     });
• }
•
•     setInterval(displayEvents, 1000);
• </script>
• </body>
• </html>

```

OUTPUT:



3) Write a program that:

- Accepts a dataset (array of numbers)
- Calculates:
 - Mean, Median, Mode
 - Standard deviation
- Display results with proper formatting

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>Statistics Calculator</title>
•
•   <style>
•     body {
•       font-family: Arial, sans-serif;
•       background: #eef2f3;
•       padding: 20px;
•     }
•
•     .container {
•       max-width: 700px;
•       margin: auto;
•       background: white;
•       padding: 20px;
•       border-radius: 10px;
•     }
•
•     input {
•       width: 100%;
•       padding: 10px;
•       margin-top: 10px;
•     }
•
•     button {
•       margin-top: 10px;
•       padding: 10px 20px;
•       background: #007bff;
•       color: white;
•       border: none;
•       cursor: pointer;
•     }
•
•     #output {
```

```

    margin-top: 20px;
  }
</style>
</head>
<body>

  <div class="container">
    <h2>Statistical Data Analyzer</h2>

    <p>Enter numbers separated by commas:</p>

    <input type="text" id="numbers" placeholder="10,20,30,40,50">

    <button onclick="calculateStats()">Calculate</button>

    <div id="output"></div>
  </div>

  <script>
    function calculateStats() {
      let input = document.getElementById('numbers').value;

      let data = input.split(',').map(Number);

      data.sort((a, b) => a - b);

      let mean = data.reduce((a, b) => a + b, 0) / data.length;

      let median;

      if(data.length % 2 === 0) {
        median = (data[data.length/2 - 1] + data[data.length/2]) /
2;
      } else {
        median = data[Math.floor(data.length/2)];
      }

      let frequency = {};
      let maxFreq = 0;
      let mode;

      data.forEach(num => {
        frequency[num] = (frequency[num] || 0) + 1;

        if(frequency[num] > maxFreq) {

```

```

    •         maxFreq = frequency[num];
    •         mode = num;
    •     }
    • });
    •
    •     let variance = data.reduce((sum, num) => {
    •         return sum + Math.pow(num - mean, 2);
    •     }, 0) / data.length;
    •
    •     let stdDeviation = Math.sqrt(variance);
    •
    •     document.getElementById('output').innerHTML = `
    •         <h3>Results</h3>
    •         <p><strong>Mean:</strong> ${mean.toFixed(2)}</p>
    •         <p><strong>Median:</strong> ${median}</p>
    •         <p><strong>Mode:</strong> ${mode}</p>
    •         <p><strong>Standard Deviation:</strong>
    •     ${stdDeviation.toFixed(2)}</p>
    •     `;
    •     }
    • </script>
    • </body>
    • </html>

```

OUTPUT:

Statistics Calculator

Ask Gemini

127.0.0.1:5500/Task2/statistics_calculator.html

Statistical Data Analyzer

Enter numbers separated by commas:

Calculate

Results

Mean: 94678.50

Median: 39.5

Mode: 4

Standard Deviation: 211626.17

4) Design an EMI calculator:

- Input: Principal, Rate, Time
- Output:
 - Monthly EMI
 - Full amortization schedule (month-wise breakdown)
- Use appropriate mathematical formulas

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>EMI Calculator</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f4f4;
      padding: 20px;
    }

    .container {
      max-width: 900px;
      margin: auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
    }

    input {
      width: 100%;
      padding: 10px;
      margin-top: 10px;
    }

    button {
      margin-top: 10px;
      padding: 10px 20px;
      background: #28a745;
      color: white;
      border: none;
```

```

        cursor: pointer;
    }

    table {
        width: 100%;
        border-collapse: collapse;
        margin-top: 20px;
    }

    table, th, td {
        border: 1px solid #ccc;
    }

    th, td {
        padding: 10px;
        text-align: center;
    }
</style>
</head>
<body>

    <div class="container">
        <h2>EMI Calculator</h2>

        <input type="number" id="principal" placeholder="Principal Amount">
        <input type="number" id="rate" placeholder="Annual Interest Rate (%)">
        <input type="number" id="time" placeholder="Loan Duration (Years)">

        <button onclick="calculateEMI()">Calculate EMI</button>

        <div id="result"></div>

        <table id="scheduleTable">
            <thead>
                <tr>
                    <th>Month</th>
                    <th>EMI</th>
                    <th>Interest</th>
                    <th>Principal</th>
                    <th>Balance</th>
                </tr>
            </thead>
            <tbody></tbody>
        </table>
    </div>

```

```

<script>
    function calculateEMI() {
        let P = parseFloat(document.getElementById('principal').value);
        let annualRate = parseFloat(document.getElementById('rate').value);
        let years = parseFloat(document.getElementById('time').value);

        let R = annualRate / 12 / 100;
        let N = years * 12;

        let EMI = (P * R * Math.pow(1 + R, N)) /
            (Math.pow(1 + R, N) - 1);

        document.getElementById('result').innerHTML = `
            <h3>Monthly EMI: ₹${EMI.toFixed(2)}</h3>
        `;

        let balance = P;
        let tbody = document.querySelector('#scheduleTable tbody');
        tbody.innerHTML = '';

        for(let month = 1; month <= N; month++) {
            let interest = balance * R;
            let principal = EMI - interest;
            balance -= principal;

            let row = `
                <tr>
                    <td>${month}</td>
                    <td>₹${EMI.toFixed(2)}</td>
                    <td>₹${interest.toFixed(2)}</td>
                    <td>₹${principal.toFixed(2)}</td>
                    <td>₹${Math.abs(balance).toFixed(2)}</td>
                </tr>
            `;

            tbody.innerHTML += row;
        }
    }
</script>
</body>
</html>

```


OUTPUT:

EMI Calculator

Ask Gemini

127.0.0.1:5500/task2/emi_calculator.html

EMI Calculator

1000

12

5

Calculate EMI

Monthly EMI: ₹22.24

Month	EMI	Interest	Principal	Balance
1	₹22.24	₹10.00	₹12.24	₹987.76
2	₹22.24	₹9.88	₹12.37	₹975.39
3	₹22.24	₹9.75	₹12.49	₹962.90
4	₹22.24	₹9.63	₹12.62	₹950.28
5	₹22.24	₹9.50	₹12.74	₹937.54
6	₹22.24	₹9.38	₹12.87	₹924.67
7	₹22.24	₹9.25	₹13.00	₹911.67
8	₹22.24	₹9.12	₹13.13	₹898.55